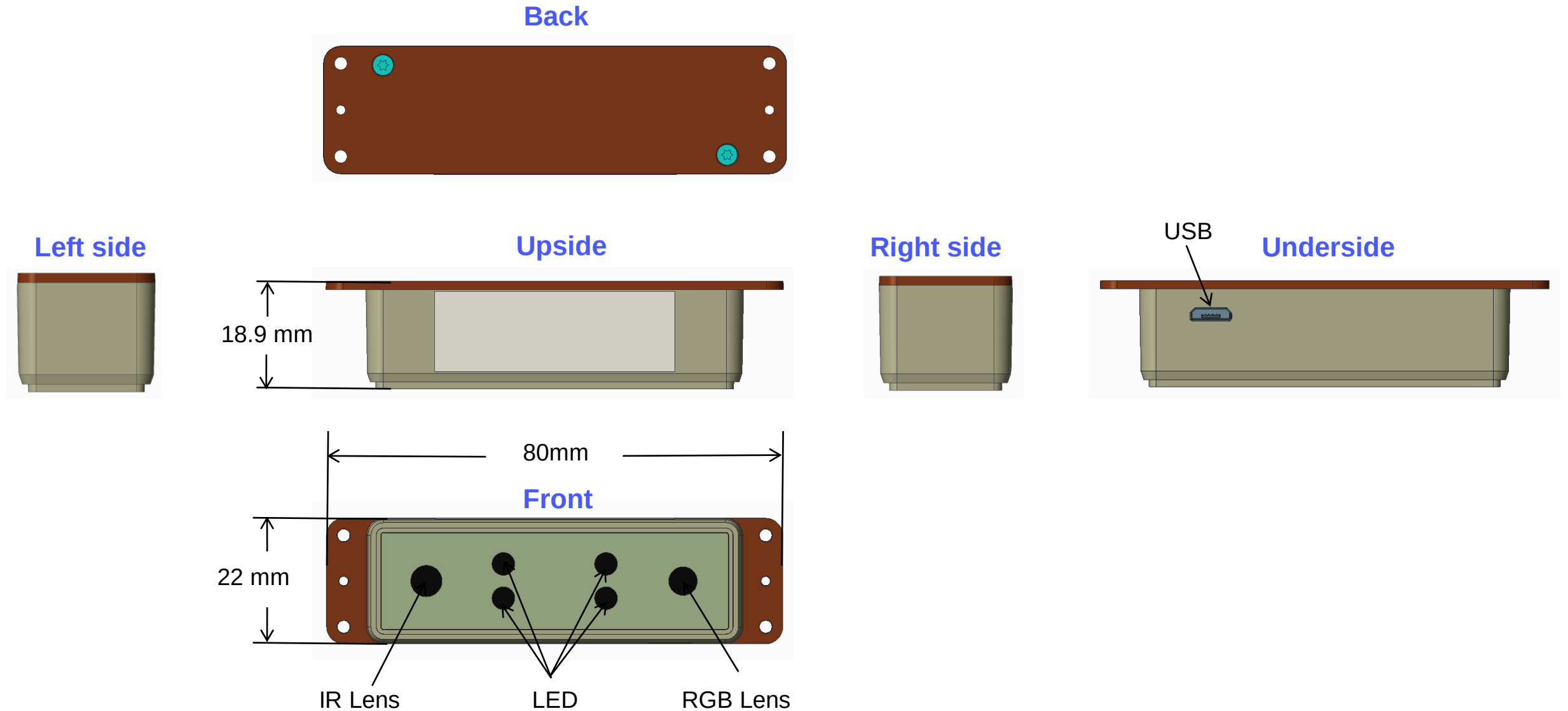


Installation and Thermal Design Guideline

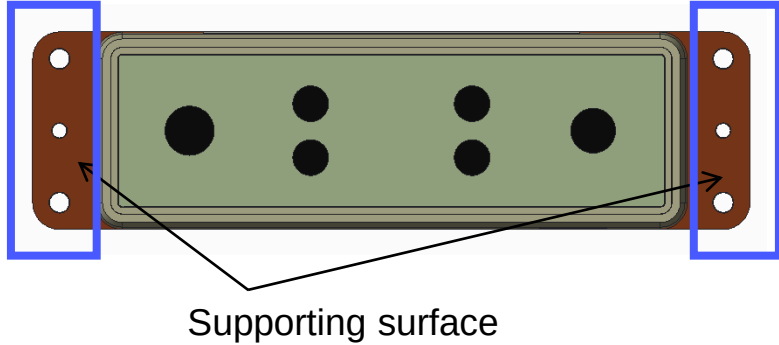
SenseNebula AIM M30 AI Module

1. Outline Overview

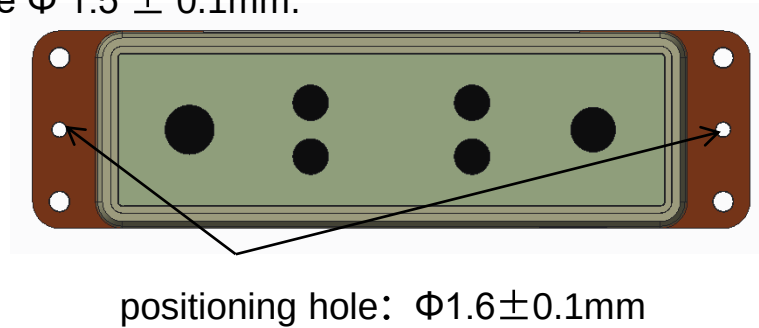


2. Installation Guidelines

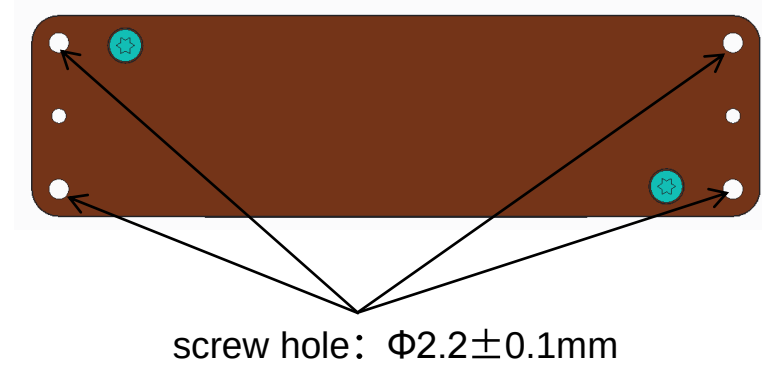
- 1) The front undershell blue frame area can be used as the mounting support surface.



- 2) Hole $\Phi 1.6 \pm 0.1\text{mm}$ can be used as an installation positioning hole to fit with the positioning pin of the entire machine. It is recommended that the positioning pin size $\Phi 1.5 \pm 0.1\text{mm}$.

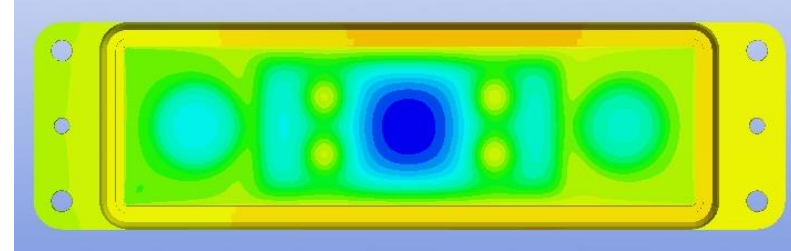
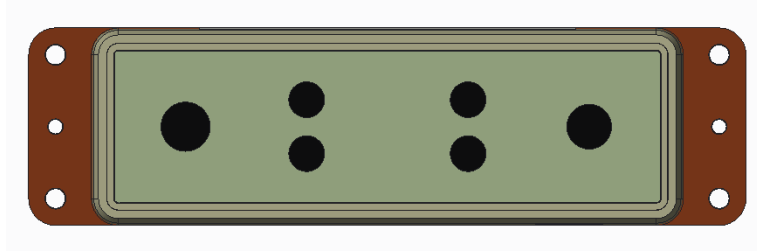


- 3) Hole $\Phi 2.2 \pm 0.1\text{mm}$ can be used as a screw hole for fastening the module and the entire machine. The recommended screw size is M2 * P0.4mm.

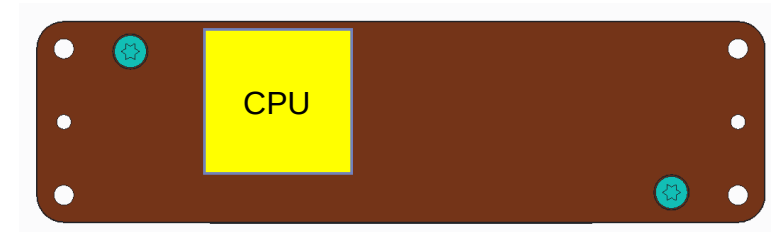
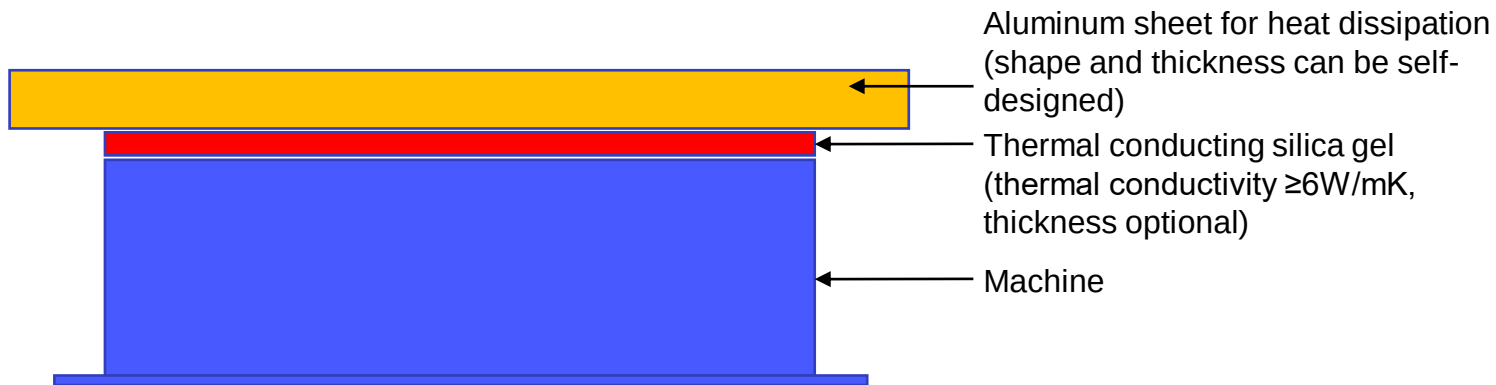


3. Thermal Design

- 1) The front Cover Glass LED position is the main heat source, and the lens area should maintain air circulation as much as possible to dissipate heat through thermal convection.

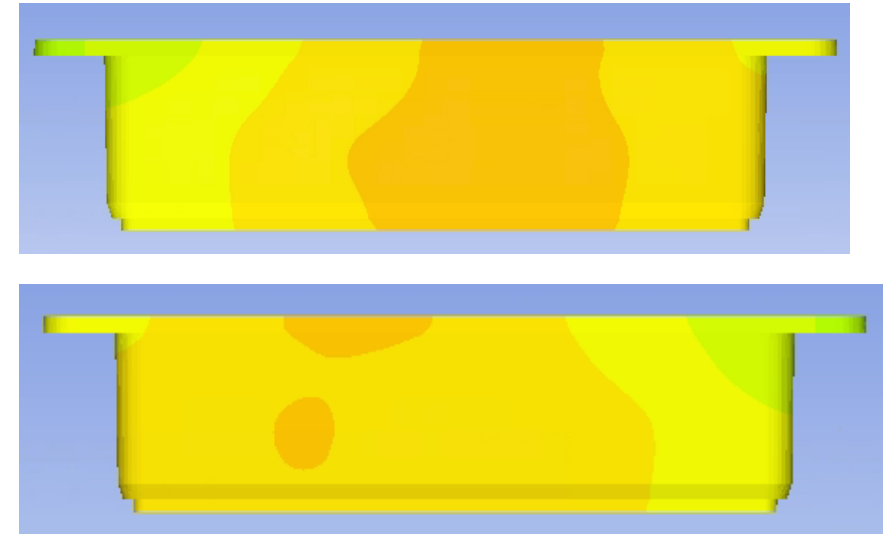
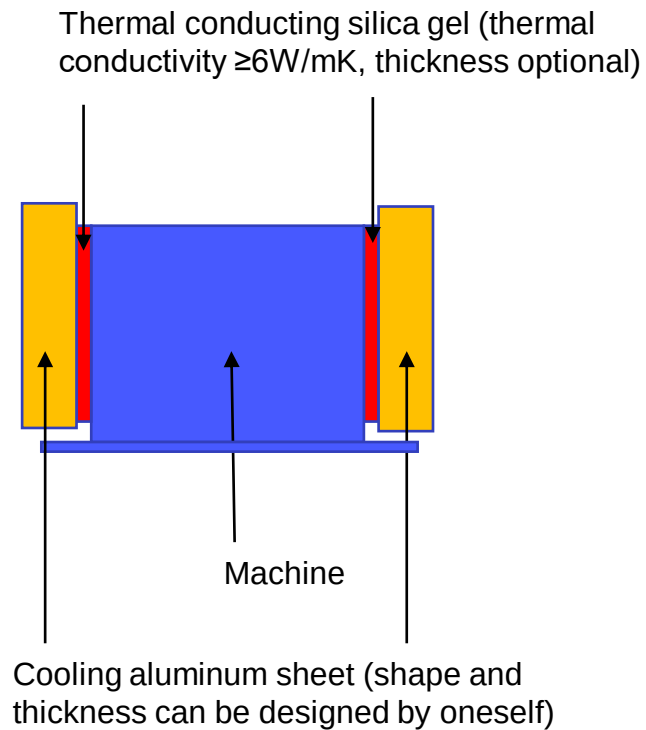


- 2) The bottom CPU is the main heat source, with a size of 14 * 14mm. Heat dissipation treatment is necessary. It is recommended to apply heat dissipation silicone (with a thermal conductivity of $\geq 6\text{W/mK}$) to the bottom and use aluminum sheets to compress and dissipate heat. The shape of the aluminum sheet can be designed according to needs, and the volume of the aluminum sheet is recommended not to be less than 30000mm^3 .



4. Thermal Design guidelines

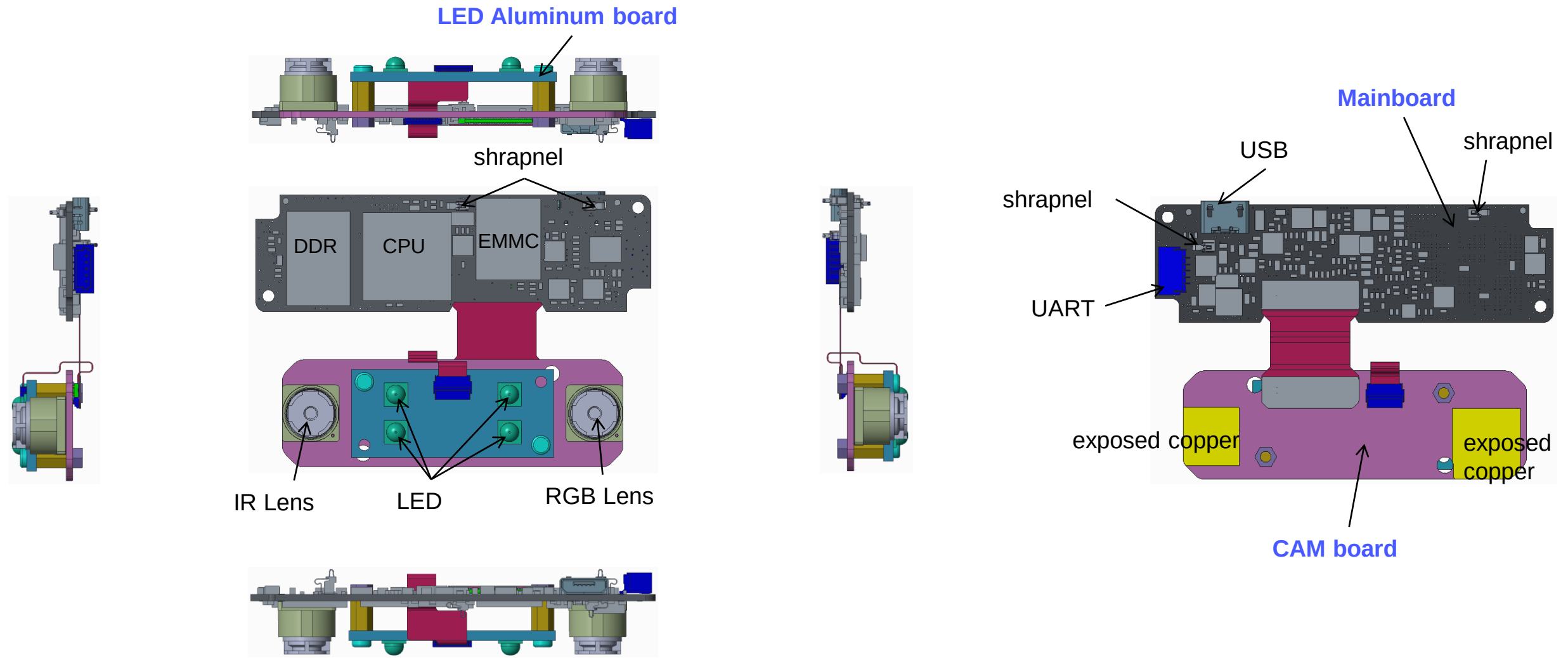
3) If conditions permit, heat dissipation treatment can also be carried out on the side of the machine. It is recommended to apply heat dissipation silicone (with a thermal conductivity of $\geq 6\text{W/mK}$) on two large sides and use aluminum sheets to compress and dissipate heat. The shape of the aluminum sheets can be designed according to needs, and the volume of the aluminum sheets is recommended to be no less than 20000mm^3 .



Installation and Thermal Design Guideline

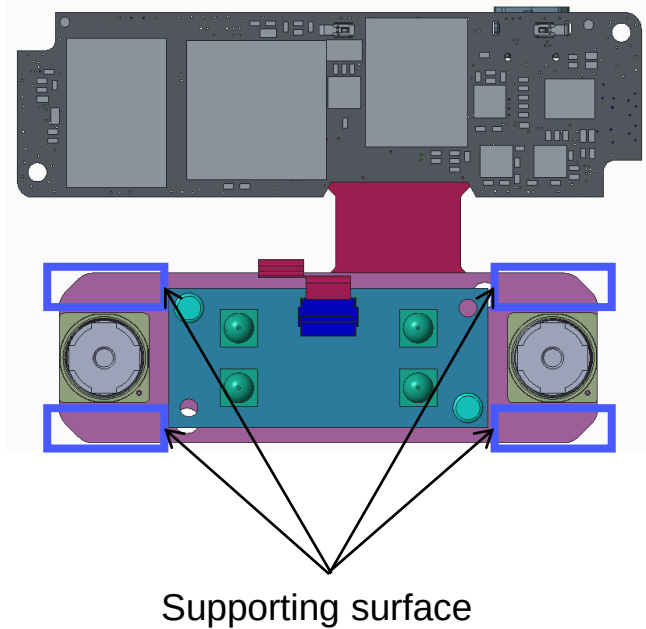
SenseNebula AIM M30 AI Module (without shell)

1. Outline Overview

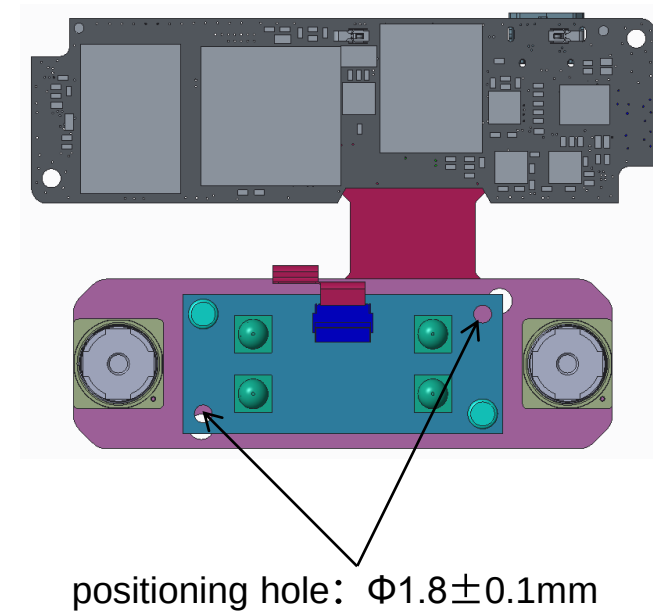


2. Installation Guidelines

- 1) The blue frame area of the CAM board can serve as the installation support surface.

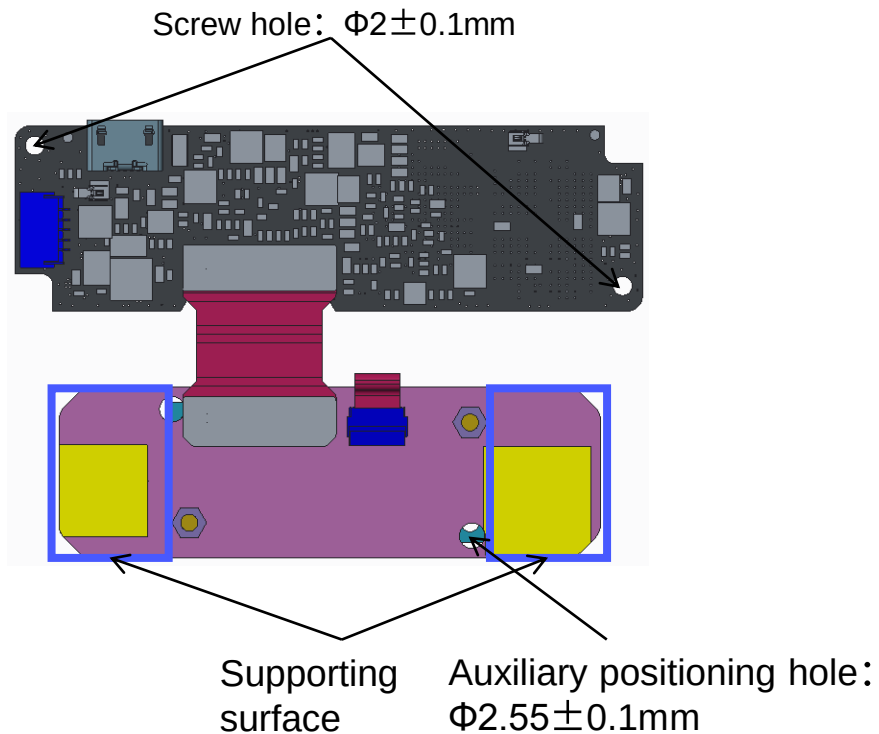


- 2) LED aluminum board $\Phi 1.8 \pm 0.1\text{mm}$ can be used as an installation positioning hole to fit with the positioning pin of the entire machine. It is recommended that the positioning pin size is $\Phi 1.7 \pm 0.1\text{mm}$.



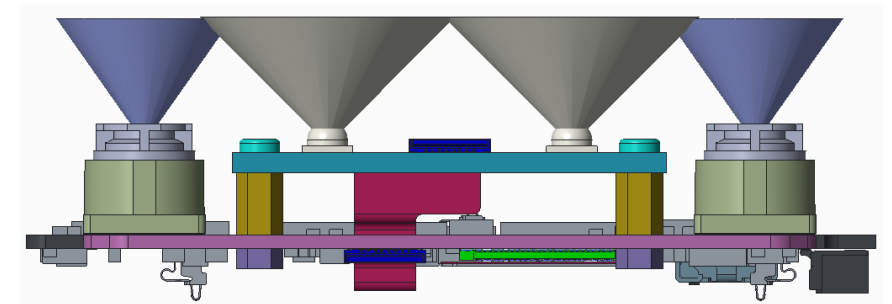
3. Installation Guidelines

- 3) ① The blue frame area of the CAM PCB can be used as a mounting and pressing plane, and can be fixed by pressing plates or other fasteners;
- ② Auxiliary positioning hole $\Phi 2.55 \pm 0.1\text{mm}$ can be used as a positioning hole for the pressure plate, which is compatible with the positioning pin of the pressure plate. It is recommended to size the positioning pin $\Phi 2.45 \pm 0.1\text{mm}$;
- ③ Hole $\Phi 2 \pm 0.1\text{mm}$ can be used as a screw hole for fastening the motherboard and the entire machine. The recommended screw size is M1.7 * P0.35mm.



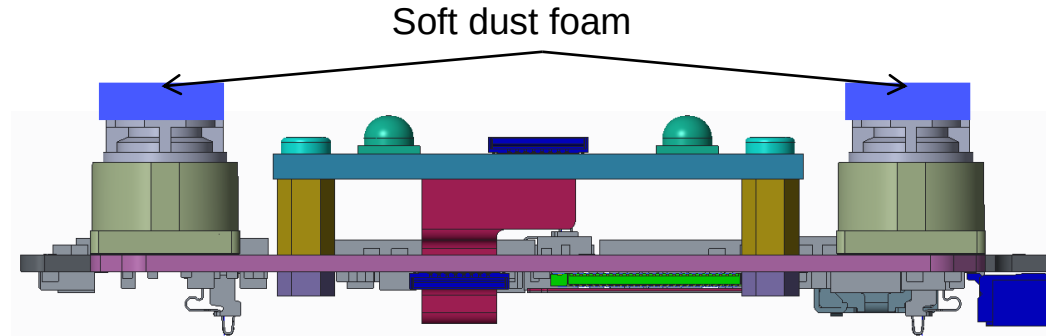
- 4) Lens/LED lights should pay attention to the overall window size to prevent hidden corners or poor recognition caused by mechanism obstruction.

- ① Lens DFOV $70.4^\circ \pm 3^\circ$;
- ② LED FOV 90° , can be partially occluded, recommended angle should be $>70.4^\circ$.

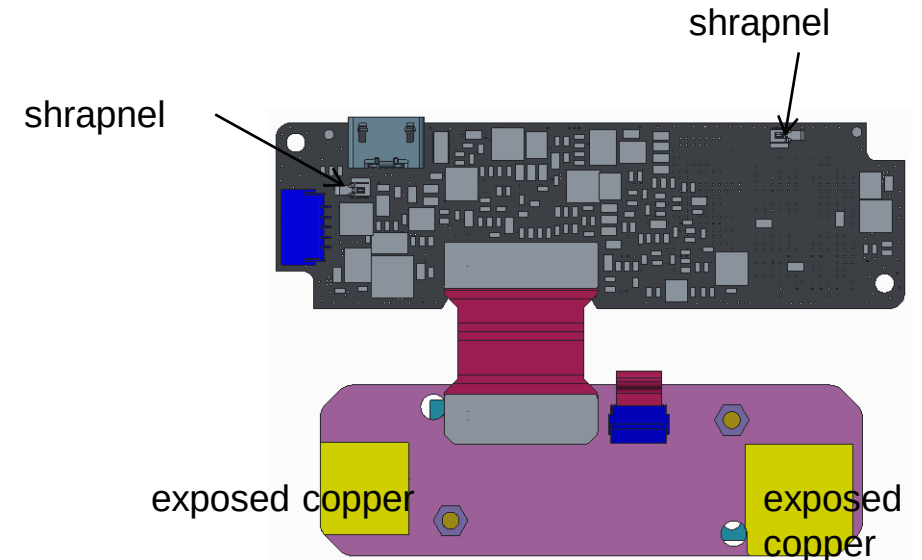
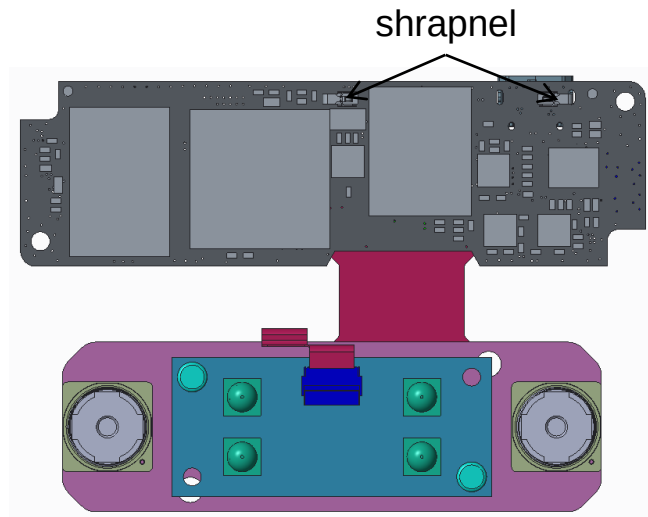


4. Installation Guidelines

- 5) It is recommended to do dustproof treatment on the Lens end face, for example, the Lens end face can be attached with soft dustproof foam. It should be noted that too much foam interference can cause Lens to compress and run away. It is recommended to have an interference of about 0.1mm.

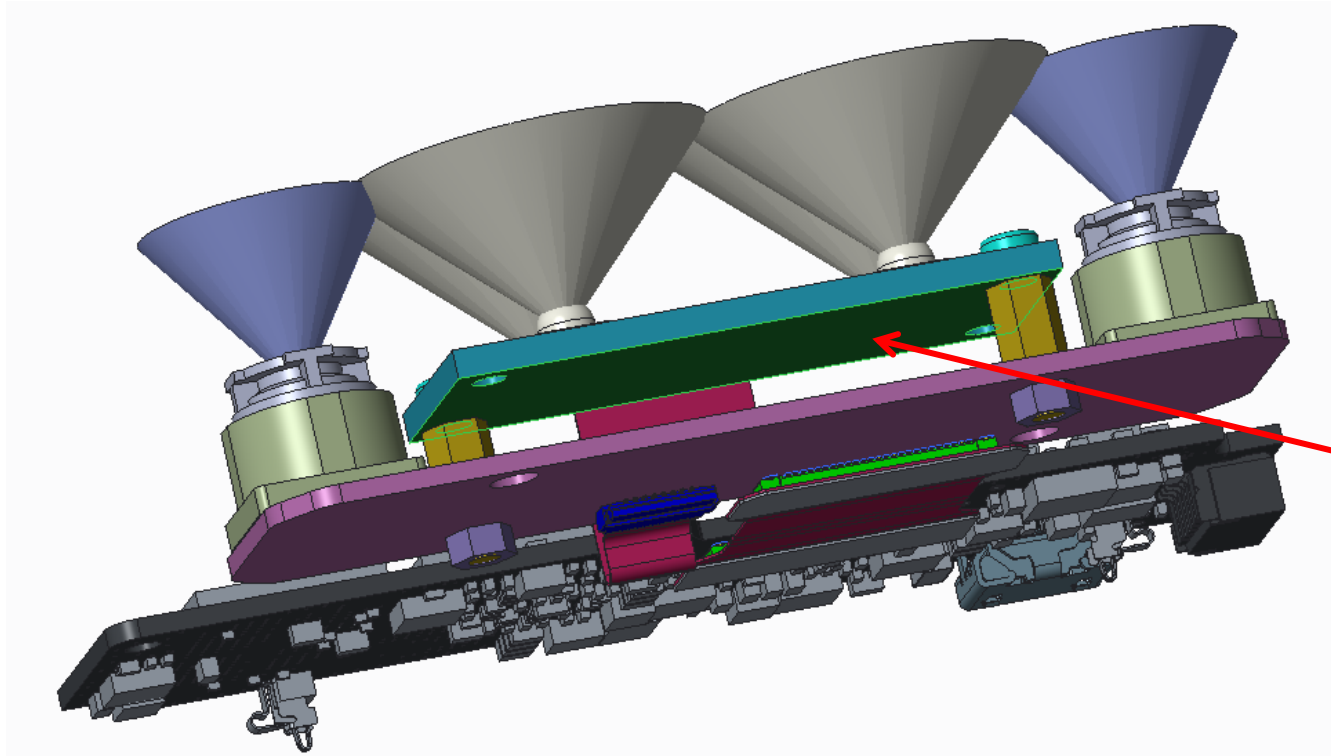


- 6) Ground the shrapnel position of the motherboard and the exposed copper position of the CAM board.



5. Thermal Design guidelines

- 1) The LED on the aluminum board is the main heat source and **must** be subjected to heat dissipation treatment. Thermal conductive silicone (with a thermal conductivity of $\geq 6\text{W/mK}$) can be attached to the bottom surface as shown in the figure and compressed with aluminum blocks for heat dissipation. The recommended area is $32 \times 14\text{mm}$.



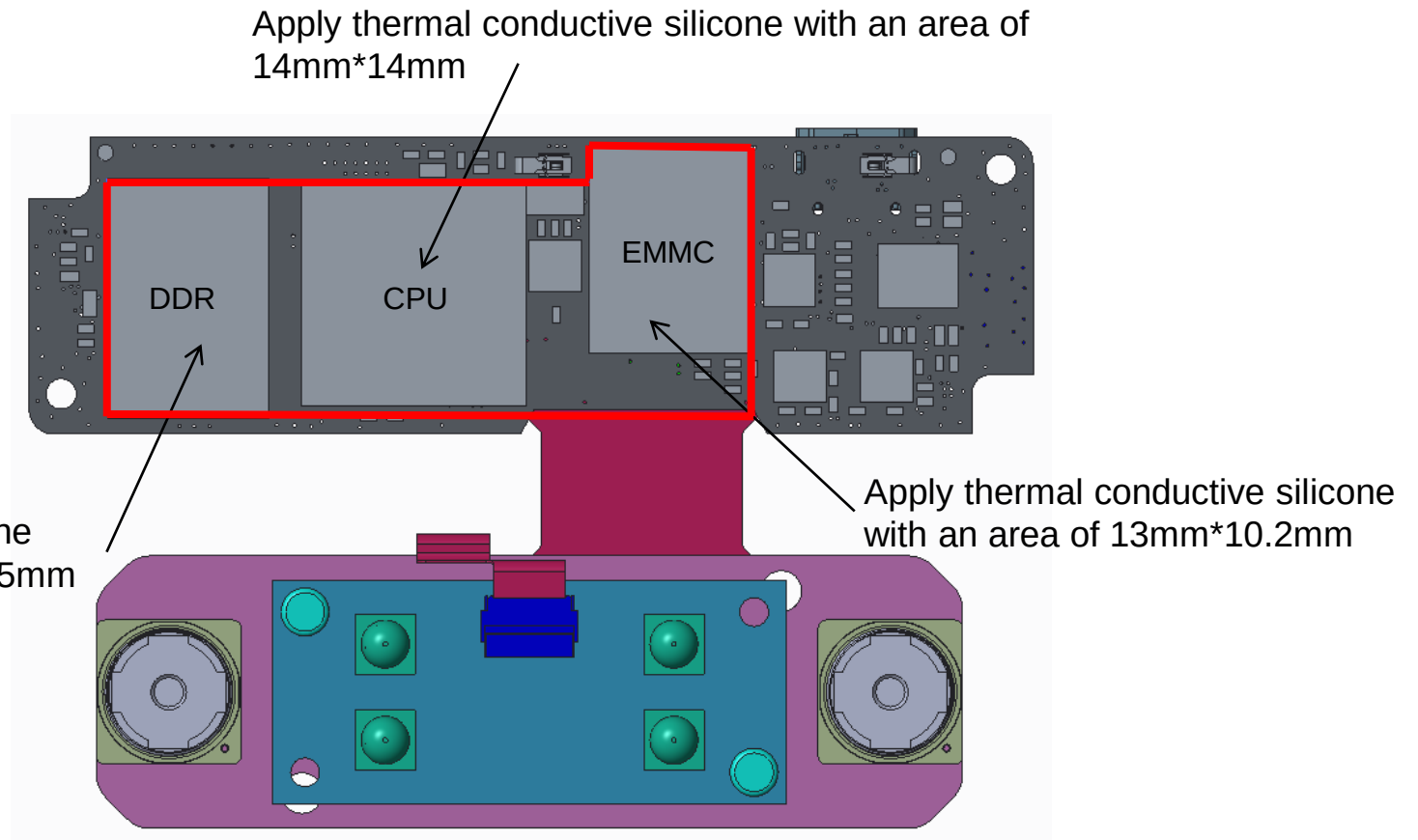
This surface paste thermal silicone,
thermal conductivity $\geq 6\text{W/mK}$

6. Thermal Design guidelines

2) The CPU/EMMC/DDR on the mainboard is the main heat source and **must** undergo heat dissipation treatment. Thermal conductive silicone (with a thermal conductivity of $\geq 6\text{W/mK}$) can be attached to the CPU/EMMC/DDR, and aluminum blocks can be used to compress and dissipate heat.

Solution 1: Apply one piece of thermal conductive silicone separately according to the diagram and use aluminum blocks to compress and dissipate heat;

Solution 2: Stick a large piece of thermal conductive silicone as shown in the diagram and use an aluminum block to compress and dissipate heat;



Remarks: the volume of the aluminum sheet is recommended not to be less than 30000mm^3